

4. Write a python program to solve vacuum cleaner problem

Aim:

To write a python program to solve vacuum cleaner problem .

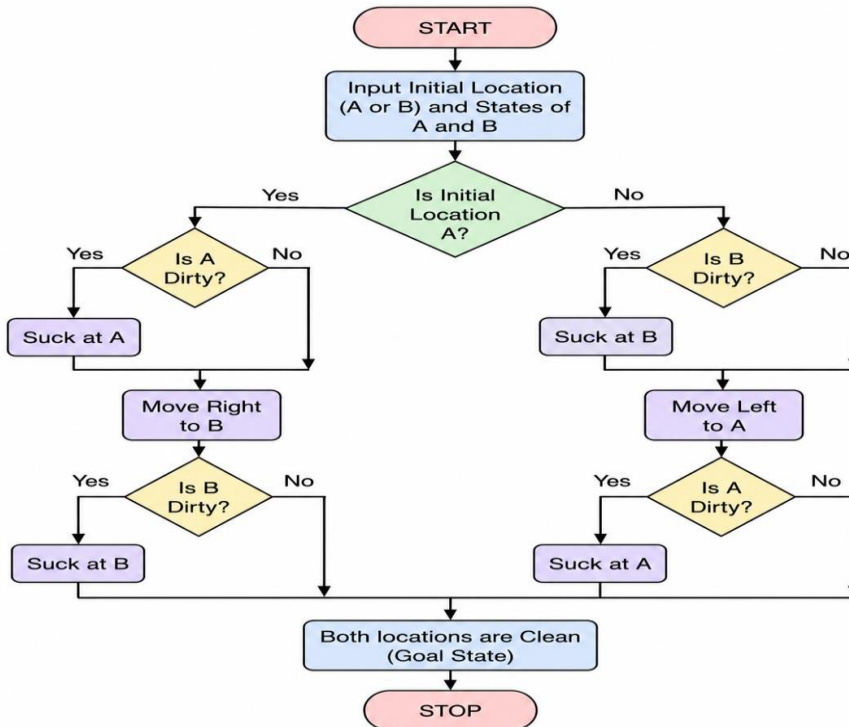
Objectives

The objective of the Vacuum Cleaner Problem is to design an intelligent agent that can sense whether a location is clean or dirty and perform the appropriate action to clean it. The agent uses actions such as Suck, Move Left, and Move Right to clean all dirty locations efficiently. The main goal is to ensure that the entire environment becomes clean using the minimum number of actions.

Algorithm :

1. Start.
2. Place the vacuum cleaner at the initial location (A or B).
3. Check whether the current location is Dirty or Clean.
4. If the location is Dirty, perform the Suck operation.
5. Move to the other location.
6. Check whether the new location is Dirty or Clean.
7. If the location is Dirty, perform the Suck operation.
8. Repeat until both locations are Clean.
9. Stop.

Flow chart :



Source Code :

```

location = input().upper()
A = input().capitalize()
B = input().capitalize()

```

```

if location == "A":
    if A == "Dirty":
        A = "Clean"
        print("Suck")
    print("Right")
    if B == "Dirty":
        B = "Clean"
        print("Suck")

```

```

elif location == "B":
    if B == "Dirty":
        B = "Clean"
        print("Suck")
    print("Left")
    if A == "Dirty":
        A = "Clean"
        print("Suck")

```

```
print(A)  
print(B)
```

Result :

The Vacuum Cleaner Problem was successfully implemented. The vacuum cleaner agent cleaned all dirty locations and reached the goal state where both locations are clean.

Output :

main.py



Run

Output

Clear

```
1 location = "A"
2 A = "Dirty"
3 B = "Dirty"
4
5 if location == "A":
6     if A == "Dirty":
7         print("Suck at A")
8         A = "Clean"
9
10    print("Move Right")
11
12    if B == "Dirty":
13        print("Suck at B")
14        B = "Clean"
15
16 else:
17     if B == "Dirty":
18         print("Suck at B")
19         B = "Clean"
20
21    print("Move Left")
22
23    if A == "Dirty":
24        print("Suck at A")
25        A = "Clean"
26
27 print("Goal Achieved: Both locations are clean")
```

Suck at A
Move Right
Suck at B
Goal Achieved: Both locations are clean

=== Code Execution Successful ===

